Row-Level Structural Integrity Diagnostics for Synthetic Tabular Data

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Abstract

Evaluating synthetic tabular data requires auditing record-level integrity beyond global distribution alignment. Standard statistical metrics assess column distributions in aggregate yet miss row-level conditional dependency violations. We present the Hybrid Integrity Framework (HIF), a row-level diagnostic composing a Logical Sentinel Ensemble (LSE) for categorical dependencies, Neighbor-Invariant Continuity (NIC) for continuous consistency, and mined implication rules for hard constraints. We make two contributions: (1) **Robust Structural Diagnostics**: across four generators (Gaussian Copula, Vine Copula, CTGAN, TVAE) and five datasets, HIF detects row-level dependency ruptures even where aggregate metrics report near-perfect fidelity (KS up to $\approx$ **98%**); and (2) **Label-Conditional Curation**: pruning low-HIF records ($\boldsymbol{H < 0.5}$) improves downstream utility when the audited structure includes the prediction target (**+7.7** points F1 on Census ACS CTGAN, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.002}$; **+20.8** points for the Vine Copula, $\boldsymbol{p < 0.001}$), and is utility-neutral otherwise (Gaussian Copula, $\boldsymbol{p = 0.90}$). Two experiments confirm: across nine Census ACS targets, nine of ten hub-target pairs recover significantly (sole exception: CTGAN×tenure, $\boldsymbol{p = 0.109}$), while one of eight non-hub pairs does on identical retained cohorts (Vine Copula×cost_burden_pct, $\boldsymbol{p = 0.016}$); withholding the target from hub selection eliminates the gain. Filtering therefore removes records whose target is improbable given features, not generic structural repairs. HIF provides a row-level integrity signal with a checkable precondition for when filtering pays off.

Keywords: Synthetic tabular data; Data integrity; Row-level diagnostics; Dependency auditing; Conditional dependencies; Generative models

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN [4], and the refinement of flexible statistical models, such as Vine Copulas [5], these models have matched aggregate marginal distributions closely. However, a critical “Dependency Gap” remains: while these models excel at replicating per-column distributions, they frequently produce records that are individually plausible under each column’s marginal yet violate the conditional inter-feature dependencies of the real-world joint distribution [6]—violations invisible to any metric that evaluates columns in isolation. Figure 1 illustrates the gap on the Supermarket Sales dataset, where a Gaussian Copula reproduces the 1D marginals while breaking the dataset’s arithmetic identity at the record level.

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying marginal distributional alignment, these metrics are inherently *aggregation-blind*: they evaluate each column’s distribution independently and cannot detect records that violate conditional inter-feature dependencies yet remain within the expected marginal range of each individual column. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights, but they yield a dataset-level attribution summary rather than per-record joint-conditional scores and scale exponentially with the feature count (Appendix D.3). Crucially, this pathology is not exclusive to the opacity of deep neural networks. As a designed positive control we include a Vine Copula arm whose categorical columns are sampled independently from their marginal distributions—its categorical–numeric dependence is deliberately not modelled—and it reproduces the pattern exactly: near-perfect aggregate marginal fits alongside pervasive row-level dependency violations.

To address these challenges, we present the Hybrid Integrity Framework (HIF), a row-level diagnostic system for detecting logical inconsistencies in synthetic tabular data. HIF is an integrative design that composes established building blocks—conditional classifiers, quantile-calibrated confidence thresholds, spectral embeddings, and mined association rules—into a single non-compensatory audit; its contribution lies in the problem formulation (per-record joint-conditional plausibility with record-level attribution) and in the systematic validation of when such auditing adds utility. HIF recasts fidelity evaluation as a multidimensional consistency detection problem, moving beyond aggregate statistical snapshots to identify logical violations at the record level and enabling the curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic; Section 3 formalizes the methodology, with calibration and auditing algorithms detailed in the Appendix.

Our framework comprises three complementary auditing layers:

1. **Logical Sentinel Ensemble (LSE):** An array of discriminative classifiers that identify high-dependency feature hubs—categorical columns or continuous features reduced to quantile bins—in the ground-truth data using cross-validated classification accuracy.
2. **Neighbor-Invariant Continuity (NIC):** A continuous feature auditing mechanism that verifies local semantic adjacency within SVD-embedded categorical contexts using robust percentile thresholding.
3. **Automated Rule Extractor:** An association rule mining engine [10] that extracts high-confidence deterministic implication rules ($A \implies B$) from ground-truth data to enforce hard structural constraints.

Summary of Contributions.

HIF contributes (i) an **Integrative Row-Level Structural Diagnostic**: by composing predictive-synergy hub selection, quantile-calibrated conditional classifiers, robust spectral-continuity auditing, and mined implication rules into one framework, it reframes synthetic data auditing from aggregate column matching to record-level conditional plausibility with non-compensatory multiplicative aggregation ($\prod_l (1-\mathcal{P}_l)$) to detect localized structural hallucinations ("impossible junctions") where every individual column value appears marginal-plausible yet their joint combination violates conditional logic. The distinctive claim is the constraint-mining bridge: HIF discovers its audit constraints automatically—implication rules via the Apriori engine and conditional classifiers via predictive-synergy hub selection—rather than taking them as externally specified input, and to our knowledge it is the first framework to apply functional-dependency and denial-constraint discovery to per-record auditing of synthetic tables (Section 2); and (ii) **Label-Conditional Curation**: we characterize when integrity filtering improves downstream utility and identify the mechanism that governs it. Pruning low-HIF records ($H < 0.5$) yields statistically significant gains on structural-error cohorts (+7.7 points F1 for CTGAN on Census ACS, paired t-test $p = 0.002$, 95% CI $[+0.038, +0.116]$; +20.8 points for the Vine Copula, $p < 0.001$, CI $[+0.187, +0.229]$), while on largely error-free cohorts (e.g., Gaussian Copula on Census ACS) filtering preserves utility without detectable degradation ($p = 0.90$). Two controlled experiments (Section 4.3) establish that this gain is *label-conditional*: it materializes when the auditor's selected hubs include the downstream target and vanishes when they do not, whether that is varied by changing the target or by withholding the target from hub selection. Integrity filtering in these settings therefore acts as conditional label curation—removing records whose target value is improbable given their features—rather than as generic structural repair. This sharpens the deployment guidance the result supports (Section 5.2): the precondition is checkable from the auditor's hub set alone, before any downstream label is used.

By providing a scalable, data-driven diagnostic, HIF enables the production of synthetic data whose structural integrity is audited for high-stakes deployment. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

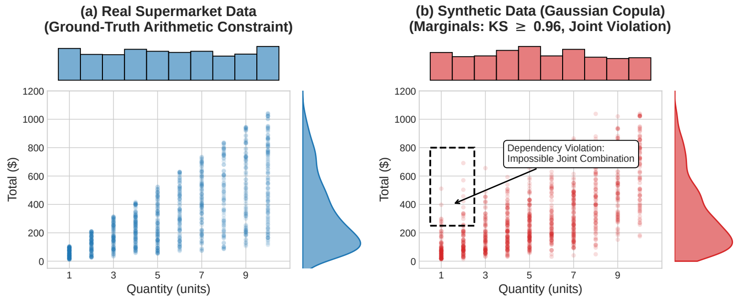

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Gaussian Copula) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms; per-column KS ≈ 0.96 for these two marginals, versus the dataset-level KS of 0.903 ± 0.006 reported in Table 1), they fail to preserve the joint arithmetic constraint ($\text{Total} = \text{Price} \times \text{Quantity} \times (1 + \text{Tax})$), hallucinating impossible records (dashed box) that aggregate marginal metrics do not capture.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models such as CTGAN and TVAE [4] set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [11], later improved for skewed distributions by CTAB-GAN [12]. Diffusion paradigms such as TabDDPM [13], flow-based gradient-boosted tree generators [14], and composable architectures such as CoDi [15] have pushed the frontier of manifold matching; permutation-invariant objectives [16] and federated frameworks [17] extend synthesis to column-order robustness and decentralized settings. However, both statistical and deep architectures suffer a fundamental objective mismatch: statistical models excel at per-column marginal fitting but smooth over complex non-linear conditional constraints, while neural models (such as CTGAN) can learn non-linear categorical contexts but remain prone to mode collapse or localized joint violations. Our work focuses on a representative spectrum of generators—from classical *Gaussian Copulas* [18] and *Vine Copulas* [5] to neural architectures such as *CTGAN*—and shows that the Dependency Gap recurs across these diverse generative paradigms in our tested settings.

Evaluation Metrics and the Dependency Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Joint Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics evaluate aggregate column-level statistics and are therefore blind to row-level conditional dependency violations—by design, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. (2025) [6] were among the first to formalize these "Dependency Gaps," proposing rule-based audits for inter-column relationships. Umesh et al. [19] studied the preservation of logical and functional dependencies from a generation-side perspective, constraining the generative model to respect mined dependencies at training time; this is complementary to—but distinct from—a post-hoc audit of an arbitrary generator's output, which is the setting we address, and dependency-aware representations have recently been shown to enhance tabular understanding [20]. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance, an explainability-aware fidelity metric, but it yields a dataset-level attribution summary rather than per-record joint-conditional scores and scales exponentially with the feature count (see Appendix D.3).

Sample-Complexity Limits on Aggregate Auditing.

Distribution-level auditing additionally faces provable sample-complexity limits: identity and uniformity tests require sample sizes that scale with the domain support [21–23], a cost only partially mitigated even for product-structured distributions [24], and conditional independence testing is likewise sample-intensive in high dimensions [25]. These bounds constrain any aggregate audit of large categorical spaces and motivate scoring records individually rather than testing the synthetic cohort globally. They are also computable at HIF's own operating configuration (Section 3): inverting the identity-testing requirement $m \gtrsim \sqrt{S}/\varepsilon^2$ [21, 23] at the per-hub, per-category sample sizes HIF actually conditions on (the datasets' actual cohort sizes, $n \in \{664, 1000, 2000\}$, over 7–24 features, top-5 hubs) yields per-cell detectability floors of $\varepsilon \approx 0.27$–1.0 across our five benchmarks versus a cohort-level analogue of 0.15–0.19 (Appendix D.2). This formalizes the granularity below which an individual sentinel cell cannot certify deviations and motivates the confidence-floor safety minimum and the multi-hub geometric aggregation of Section 3, which dampen the long-tail cells that drive the worst-case floor.

Record-Level Auditing of Synthetic Data.

A complementary line of work scores individual records rather than the synthetic cohort as a whole. Alaa et al. [26] introduced the sample-level support metrics α-Precision and β-Recall used in Section 4.1, which audit whether a record falls within the support of the real distribution in a learned embedding space; this is a neighborhood-membership notion rather than a check of conditional inter-feature logic, and the embedding is hand-selected rather than derived from the data's dependency structure. Data-SUITE [27] identifies in-distribution incongruous examples through per-feature conformal prediction intervals, but it scores each feature marginally, targets real (not synthetic) samples, and does not evaluate the joint conditional configuration of a record. DOMIAS [28] performs per-record auditing of synthetic data via membership inference against overfitted generators; this is a privacy signal—it flags records the generator memorized—rather than a general structural-integrity check. Two recent works are closest to ours and merit explicit distinction. TabStruct [29] benchmarks structural fidelity—compliance of the synthetic cohort with the causal structure of the real data—across 13 generators and 29 datasets, but its scoring is dataset-level and requires (or approximates) the ground-truth causal structure, whereas HIF scores each record against conditional dependencies mined from the data itself, including implication rules discovered Apriori-style rather than hand-specified. Curated LLM

[30] curates candidate synthetic rows in low-data regimes, but its curation signal is supervised, relying on the learning dynamics and confidence of a model trained on real labels; this is precisely the label-conditioned regime we isolate in Section 4.3, whereas HIF audits all conditional structure without any downstream label and derives a utility consequence only as an emergent, label-conditional property. To our knowledge, no prior work composes mined functional dependencies and implication rules into a per-record integrity audit of an arbitrary generator's output; the discovery literature for such constraints [31, 32] has been applied to schema design and data cleaning rather than synthetic-data auditing. That constraint-mining bridge is the novel contribution distinguishing HIF from this line of work.

HIF automates dependency discovery through predictive synergy scoring: unlike rule-mined audits, which are binary and require manual specification, HIF provides a soft probabilistic integrity score using non-compensatory multiplicative aggregation that prevents high-fidelity feature dimensions from masking localized dependency violations.

3 Mathematical Foundation

Terminology.

Throughout, four related but distinct terms are used. *Fidelity* denotes aggregate distributional alignment—how closely a synthetic cohort's marginal or pairwise statistics match the real data—as measured by metrics such as KS, TVD, and JCD. *Integrity* is the row-level property this framework audits: a record has high integrity if its joint configuration is consistent with the conditional dependencies learned from the real data, formalized as the HIF score $H(\mathbf{x})$. *Plausibility* is weaker and per-column: a record is plausible if each of its values falls within the expected marginal range of its feature, which a dependency-violating record can still satisfy. *Validity* refers to whether a record is actually possible under the true data-generating process; because HIF certifies consistency with a *learned* conditional manifold, a low-integrity record is best read as "structurally inconsistent with the training distribution" rather than universally invalid (Section 4).

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. Let $\mathcal{D}_{real}$ denote the original training dataset. We define the **ground-truth manifold** $\mathcal{M}_{real}$ as the support of the joint distribution $P_{real}(\mathbf{x})$—the localized subspace of valid feature configurations that occur in reality [33]. A synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ (with $\mathbf{x}_{cat}$ and $\mathbf{x}_{cont}$ the sub-vectors of categorical and continuous features) is a *dependency-violating record* if it is plausible under each marginal distribution yet falls outside the ground-truth manifold (i.e., it is highly improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the data). This distinction is central: aggregate fidelity metrics (KS, TVD, JCD) assess whether records fall within the expected marginal range of each column in isolation, whereas HIF additionally evaluates whether each record's joint configuration structurally aligns with the ground-truth manifold. Figure 2 shows how the three auditing layers compose into a single row-level score, and the complete calibration process for establishing these data-driven manifold boundaries is detailed in Algorithm 1.

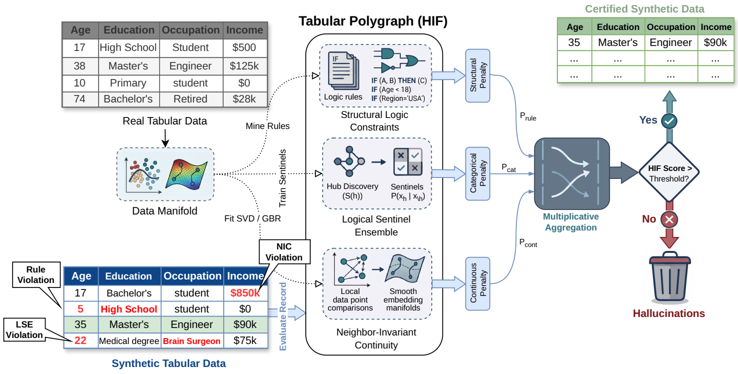

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, combined via multiplicative aggregation to filter semantically inconsistent records.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of a target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier [34]—leveraging decision tree recursive partitioning to capture non-linear feature interactions [35]—which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, the cross-validated classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x}\sim\mathcal{D}_{real}}\left[\mathbb{I}\left(\arg\max_{c\in\mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h\right)\right] \tag{1}$$

where $\mathbb{I}$ is the indicator function. By maximizing $S(h)$, we select hubs that are most strongly constrained by conditional interactions with other attributes. While computationally efficient, this reliance on single-variable classification accuracy primarily captures univariate dependencies and may under-represent higher-order parity relationships; future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection.

Consider, for illustration, a demographic dataset where x_h is *Education Level* and $\mathbf{x}_{\setminus h}$ contains *Age* (e.g., 14). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel will flag the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

Drawing inspiration from distribution-free quantile calibration in inductive conformal prediction [36–38], we establish a data-driven Confidence Floor δ_h for each hub based on empirical calibration error bounds, defined as the empirical 5th-percentile ($Q_{0.05}$) of the *out-of-bag* probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max\left(Q_{0.05}\left(\{P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\right), 0.01\right) \tag{2}$$

Using out-of-bag predictions (i.e., predictions of the trees that did not train on each row) yields out-of-sample floors without requiring a held-out split, preventing over-optimistic calibration on rows the sentinels memorized during training. This 5th-percentile floor acts as a conservative safety margin: if the real data occasionally contains 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05), and the floor δ_h ensures we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

The complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—is summarized in Algorithm 1 (Appendix B).

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a continuous latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$—drawing on classical Multiple Correspondence Analysis (MCA) and Singular Value Decomposition (SVD) on binary indicator matrices [39, 40]. Standardizing the indicator matrices by their standard deviation prior to SVD mathematically replicates the inverse-frequency weighting of MCA, ensuring that rare but highly informative categorical constraints dominate the continuous manifold projection. We then model the conditional expectation of each continuous feature y as a non-linear function of this manifold using a Gradient Boosting Regressor r_y [41, 42]:

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx r_y(\phi(\mathbf{x}_{cat})) \tag{5}$$

For any given record $\mathbf{x}$, let $e_y(\mathbf{x}) = |y(\mathbf{x}) - \hat{y}(\mathbf{x})|$ denote the absolute regression residual, and define the ground-truth residual set as $\mathcal{E}_y = \{e_y(\mathbf{x}) \mid \mathbf{x} \in \mathcal{D}_{real}\}$. For example, a synthetic 45-year-old Senior Executive generated with an Income of \$12 yields an enormous residual against the expected executive income ($\hat{y} \approx$ \$120,000), far exceeding the natural variance of executives in the real dataset and signaling a continuous dependency gap. To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y from $\mathcal{E}_y$ using the Median Absolute Deviation (MAD) [43], a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation:

$$\tau_y = \max\left(Q_{0.95}(\mathcal{E}_y), \mathrm{Median}(\mathcal{E}_y) + 2 \cdot \mathrm{MAD}(\mathcal{E}_y)\right) \tag{6}$$

This threshold keeps the NIC layer stable even in high-skew distributions, such as macroeconomic wage data.

Two scope notes bound the NIC layer's interpretation. First, the threshold conditions on the categorical context only: NIC audits each continuous feature against the learned categorical manifold $\phi(\mathbf{x}_{cat})$, so it cannot by construction detect continuous–continuous dependencies such as the arithmetic identity of Figure 1 (Total = Price × Quantity × (1 + Tax) on Supermarket Sales). Such relations are the province of the rule layer—which quantizes and mines them as implication rules—and of the LSE, which treats quantile-binned continuous features as hubs; on the Supermarket Sales audit the identity is caught by those layers while the NIC component rate is 0.0% (Section 4.1). NIC is therefore a categorical-context continuity check rather than a general multivariate residual auditor. Second, τ_y is calibrated on the residuals of the training rows themselves, and because the regressor is fitted on those same rows the in-sample residuals can modestly understate the natural noise floor. A held-out 50/50 re-calibration across our five benchmarks yields thresholds between 0.96× and 1.54× looser than the in-sample thresholds (Census ACS, where the headline analyses sit, $\leq$ 1.05×), so NIC's per-column violation rates are mild upper bounds rather than exact estimates. We retain in-sample calibration because it uses the full real cohort and the effect is second-order relative to the LSE signal; the sensitivity is disclosed rather than silently propagated.

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ uses a dynamic scaling factor $\gamma_y = \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\mathrm{MAD}(\mathcal{E}_y), 0.1)$ that bounds the penalty relative to the feature's natural noise floor, so small deviations beyond τ_y yield soft penalties while massive violations (like the \$12 executive) yield a maximum penalty of 1.0:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\mathrm{clip}\left(\frac{e_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1\right)\right] \tag{7}$$

As a defensive guard, NIC detects *collapsed* continuous features whose conditional predictions have near-zero variance (i.e., the feature is effectively constant given the categorical context); for such degenerate features, all records receive a uniform base penalty since the expected residual distribution provides no reliable baseline. This guard did not activate on any benchmark dataset in our experiments.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection: the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x})) \right)^{1/L} \tag{8}$$

where l iterates over the active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any single manifold dimension ($\mathcal{P}_l \to 1$) drives the entire record integrity toward zero, regardless of how well the other components align (see Table 7, Section 4.6). For a formal analysis of this aggregation logic under explicit assumptions—intersection soundness (Proposition 1), asymptotic convergence (Proposition 2), and the Type I false-rejection bound under quantile calibration (Remark 1)—we direct the reader to **Appendix A**.

While HIF prioritizes soft probabilistic validities via LSE and NIC, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age $\geq$ 18 for voters), bridging differentiable manifolds and rigid symbolic constraints. In our configuration, rules are mined from the real data with minimum confidence 0.95, minimum support 0.005, at most two antecedents, and up to 25 rules, and they are applied as hard binary penalties: a record violates the layer if any mined rule fails on it. Two consequences follow and are disclosed rather than left implicit. First, a rule that holds in only 95% of real records still penalizes every synthetic record in the exception class, so the layer is a hard filter rather than a probabilistic one, and the union of exception classes across up to 25 rules is material—genuine held-out real records are flagged overall at 0.0–14.0% across our benchmarks (Section 4.4). Second, component validities are floored at $\epsilon = 10^{-4}$, so a record that violates at least one mined rule attains $H(\mathbf{x}) \leq \epsilon^{1/L} \approx 0.046$ in the three-layer configuration—safely below the 0.5 frontier—even when every other layer passes it; this floor is precisely what the mode-collapse control of Section 5.3 exploits to catch collapsed transaction cohorts.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N} \sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort. The complete end-to-end auditing procedure for scoring a synthetic dataset is detailed in Algorithm 2 (Appendix B).

Because HIF penalizes records that violate *learned conditional dependencies* rather than marginal rarity—the confidence floor δ_h is calibrated on the ground-truth manifold precisely to avoid penalizing rare-but-valid tail records—filtering operates on structural consistency rather than demographic support; a systematic audit of representation drift across sensitive attributes under integrity filtering is left to future work. A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores (≥ 0.95) typical of aggregate 1D distributional metrics in standard evaluation benchmarks [2, 6, 11]. This is a deliberate design property: a record with a severe violation of any single dependency should not receive a high integrity score, regardless of how well its remaining features conform to marginal expectations. Assuming conditional independence of error dimensions given the feature representations, the product of validities $\prod(1 - \mathcal{P}_l)$ serves as a pseudo-likelihood proxy for the joint structural conformity of the record, and under empirical quantile calibration ($\delta_h = Q_\alpha$, $\gamma_y = Q_{1-\beta}$) the false-rejection probability for valid tail records is upper-bounded by the Union Bound under stated assumptions (Remark 1, Appendix A).

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on five benchmark datasets with diverse structural complexity: U.S. Census ACS [44, 45], Adult Income, Credit Card Default, Online Purchases, and Supermarket Sales. Experiments used $N = 3$–10 random seeds depending on the evaluation; we report mean $\pm$ standard deviation (SD) unless otherwise noted.

Comparison Metrics.

HIF is contrasted throughout against four aggregate fidelity metrics, all oriented so that higher is better. The **KS complement** is $1 - D$, where D is the two-sample Kolmogorov–Smirnov statistic, averaged over numeric columns; **Total Variation Distance (TVD)** is reported as its complement over categorical columns. **Moment Matching (MM)** is a per-column weighted agreement of the first four moments—mean, standard deviation, skewness, and kurtosis, with weights $(0.40, 0.35, 0.15, 0.10)$ and each discrepancy expressed as a bounded relative error—averaged over numeric columns. **Joint Correlation Distance (JCD)** compares mixed-type association matrices: for each column pair we compute Spearman $|\rho|$ (numeric–numeric), bias-corrected Cramér's V (categorical–categorical), or the correlation ratio η (mixed), assemble the matrices R_{real} and R_{syn} over d columns, and report $100 \cdot \big(1 - \|R_{\text{real}} - R_{\text{syn}}\|_F / \sqrt{d(d-1)}\big)$; despite the name, the reported quantity is thus a bounded similarity score rather than a distance. We additionally report the sample-level support-overlap metrics **α-Precision** and **β-Recall** of Alaa et al. [26], the closest prior work on row-level auditing of generative models. Our β-Recall follows their construction but thresholds against real-data hypersphere radii and omits their local-coverage factor; it is consequently less sensitive to mode collapse than the original, and should be read as a coarse support-coverage indicator rather than a reproduction of their published β-Recall.

To calibrate the framework's response, we executed a controlled semantic violation protocol: we stochastically permuted (shuffled) the values of specific features across different rows within highly constrained categorical-continuous dependencies for a controlled fraction of records $\eta \in [0, 0.6]$ (Census ACS, $N = 3$ seeds). Shuffling existing values across rows rather than injecting external noise preserves each column's marginal distribution (satisfying standard MM and KS metrics) while breaking the underlying joint conditional structure at the row level. Under this protocol, the HIF score decreased monotonically with η in every seed (per-seed averaged Spearman $\rho = -1.00$; pooled over seeds, $\rho = -0.98$—near-perfect monotonicity over a five-level sweep is nearly tautological, so the substantive evidence is the per-record attribution below): the mean HIF decreased from 0.934 ± 0.012 at $\eta = 0$ to 0.702 ± 0.013 at $\eta = 0.6$, whereas the marginal metrics—Moment Matching (MM) and Kolmogorov-Smirnov (KS)—remained static to three decimals ($\rho \approx 0$), since shuffling rearranges intact marginal values. Notably, the Joint Correlation Distance (JCD) also decayed sharply ($94.6 \rightarrow 82.6$, $\rho = -0.98$): shuffling a feature destroys its pairwise correlations, so JCD is a sensitive aggregate drift detector—but it yields only a dataset-level summary and cannot attribute degradation to specific records. This motivates HIF's row-level output and the low-rate conditional-swap protocol used in Section 4.5, whose replacements are drawn from real conditional pools rather than the intact marginals.

Evaluation Protocol and Threats to Validity.

We distinguish ranking quality (ROC-AUC, PR-AUC) from thresholded classification quality (F1), treating the former as threshold-independent diagnostics of error ranking and the latter as an operating-point metric. For baseline comparability, IF and LOF use a `contamination` parameter set to the known corruption rate, and the learned-density baseline (GMM) uses the corresponding quantile of real-data scores, in synthetic stress tests; this favors those baselines for F1 and is reported explicitly to avoid hidden calibration advantages (we likewise report HIF and GMM at a matched operating point in Section 4.4). The controlled corruption protocol above is used strictly for metric response calibration, not as the primary benchmark for diagnostic utility: our main empirical validation assesses HIF across three independent regimes—(1) a cross-architecture maturity audit of generative models (Section 4.1), (2) downstream predictive utility on un-manipulated synthetic cohorts (Section 4.2), and (3) held-out error families that HIF was not specifically engineered to detect (plus the targeted dependency-violation family as a positive control; Section 4.4). Sample sizes differ by experiment: the significance, held-out, and cross-architecture audits use $N = 10$ seeds with paired tests and exact p-values where applicable, whereas the calibration-domain experiments use $N = 3$ seeds and are reported as descriptive mean $\pm$ SD without inferential claims; the calibration-domain and statistical-significance analyses are confined to Census ACS, so their inferential conclusions should not be extrapolated to other domains without further evaluation. Residual threats include dataset selection bias and the possibility that rare-but-valid tail patterns are partially coupled with high-penalty regions. Two caveats bound the construct validity of the "violation" label. First, HIF certifies inconsistency with the *learned* conditional manifold: a flagged record need not be impossible, only sufficiently improbable under the conditional dependencies estimated from the real data, so the frontier should be read as "structurally inconsistent with the training distribution" rather than "universally invalid." Second, where invalidity is externally verifiable—the Supermarket Sales arithmetic identity (Total = Price $\times$ Quantity $\times$ (1 + Tax))—the flagged cohort was confirmed to violate the constraint, providing an independent anchor for the penalty; such verifiable constraints are the strongest evidence for the hallucination interpretation and should be prioritized when available. To bound these caveats we report, alongside every generator row in Table 1, a real-data reference row: for each dataset the auditor is fitted on real rows and scored on a genuine held-out remainder (2,000 additional rows where the cache permits; the residual 462-row remainder for Census ACS; and an 80/20 split for the 664-row and 1,000-row cohorts). This reference establishes the framework's own error floor—genuine held-out real data is flagged at 0.0–14.0% across domains—and exposes a framework-level overfitting signature: the auditor scores its own training rows higher than fresh held-out rows (e.g., Census ACS 0.985 vs. 0.964; Adult 0.977 vs. 0.923), so HIF values should be read relative to the real-data reference rather than absolutely.

4.1 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 1): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark). The results reveal a key insight—an *Integrity-Fidelity Decoupling*: a generator can match aggregate marginal statistics (KS) while simultaneously violating the joint conditional structure of the data, so that high marginal fidelity does not imply row-level integrity, and vice versa:

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, the Vine Copula arm achieves the highest marginal statistical fit of the four generators (KS $\geq$ 0.970). This arm is a designed positive control: its categorical columns are sampled independently from their marginal distributions and its categorical–numeric dependence is deliberately not modelled (as disclosed in the released implementation), and its near-perfect KS is the expected consequence of resampling the empirical marginals. It therefore maximizes the contrast between aggregate marginal fidelity and row-level joint integrity by construction. Accordingly, on datasets with cross-feature relational constraints (Supermarket Sales and Online Purchases) it exhibits severe joint dependency failures, with violation rates of **85.5% to 99.9%** (HIF $\leq$ 0.271); CTGAN and TVAE show similar deficits on transaction constraints (HIF = 0.02–0.30, violation rates 81.3%–100%). Marginal metrics (KS) report these cohorts as high-fidelity, underscoring that aggregate fidelity metrics do not capture joint dependency ruptures. A qualitative audit on Supermarket Sales (Vine Copula) confirmed the mechanism: generators match each column's marginal distribution yet violate the arithmetic identity subtotal = unit_price × quantity × (1 + tax), with HIF flagging essentially the entire cohort (violation rate 99.9% across $N = 10$ seeds; component rates 100.0% LSE, 59.3% rules, 0.0% NIC)—while KS and joint-correlation metrics reported high fidelity. This identity is externally verifiable, independent of the auditor's learned models, and anchors the construct validity of the HIF penalty in a way that learned conditional improbability alone cannot (see the threats-to-validity discussion in Section 4).

These violations are independently confirmable beyond the auditor's own learned models. Using arithmetic identities that hold to $\approx 10^{-13}$ relative error in the real data—Total = Unit Price × Quantity × 1.05 (equivalently Total = cogs + gross income) on Supermarket Sales, and item_total = item_subtotal + item_tax together with item_subtotal = purchase_price × quantity on Online Purchases—we verified every real generator cohort in this audit ($N = 10$ seeds). Records flagged by HIF ($H < 0.5$) are independently confirmed as identity-violating in $\geq 98.9\%$ of cases across all eight dataset–generator configurations (1.000 in six of eight; Table 2). This verification re-fits HIF on the projection onto the five identity-bearing columns, where the arithmetic identity becomes a near-deterministic hub, so its Flagged rate is not directly comparable to the full-feature Viol. Rate of Table 1—the projection isolates the constraint and thus flags more rows (e.g., 32.0% vs. 3.4% for Gaussian Copula on Supermarket Sales), which is precisely why it serves as the sharper external anchor for the penalty. The constraints are violated wholesale (base violation rate 0.87–1.00), so on these fully constrained domains the identity check confirms the Dependency Gap rather than discriminating among records: the cleanest discriminator is Gaussian Copula on Online Purchases, where unflagged records violate at 0.715 versus 0.989 among flagged, and the HIF penalty correlates with violation severity (Spearman $\rho = 0.46 \pm 0.01$; Table 2). Conversely, where HIF flags essentially the entire cohort (Vine and CTGAN on Supermarket Sales; Vine, CTGAN, and TVAE on Online Purchases), its penalty is only weakly monotone in arithmetic severity ($\rho \leq 0.44$), because the cohort's failures are dominated by categorical-structure violations rather than continuous residuals—precisely the regime in which the learned-density baseline of Section 4.4 is the sharper detector of the arithmetic constraint (GMM ROC-AUC 0.955 vs. 0.797 on Online Purchases). These numbers quantify the earlier qualitative audit: the flagged cohort is externally invalid at near-universal rates, while HIF's *ranking* within arithmetic-constraint domains is limited—a boundary that motivates the complementary learned-density baseline reported in Section 4.4.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.31 to 0.99 (violation rates 0.2%–70.3%), providing a continuous gradient of structural integrity for comparing generator maturity. Notably, TVAE achieves the highest HIF on both Credit Default (0.908) and Adult Income (0.990). Read against the auditor's own reference on genuine held-out real data (Table 1): Credit Default real rows score HIF 0.823 with a 14.0% violation rate, and Adult real rows score 0.923 with 4.3%. TVAE on Adult (0.990, 0.2% violations) therefore exceeds the real-data reference on both quantities—evidence of genuine joint-dependency preservation—and TVAE on Credit (0.908, 8.0%) likewise exceeds it (0.823, 14.0%), so in these runs TVAE's joint-dependency preservation generalizes beyond the auditor's own training distribution rather than reproducing its estimation-error floor. The auditor also scores its own training data higher than fresh held-out real rows (Section 4), a framework-level overfitting signature that argues for reading HIF values relative to the real-data reference rather than absolutely.

3. Quantitative Decoupling Evidence.

Pearson correlation analysis at the per-seed level across the four audit benchmarks ($n = 160$: 16 dataset–generator configurations $\times$ 10 seeds, with Census ACS held out as the calibration benchmark; Table 1) characterizes the decoupling precisely. Seeds within a configuration are not independent observations, so these correlations are descriptive rather than inferential: HIF shows no linear relationship with marginal fidelity ($\rho(\mathrm{HIF}, \mathrm{KS}) = 0.094$) or support coverage ($\rho(\mathrm{HIF}, \alpha) = 0.06$), but a strong positive correlation with joint correlation fidelity ($\rho(\mathrm{HIF}, \mathrm{JCD}) = +0.83$). HIF thus aligns with joint-dependency structure while being decoupled from per-column marginal metrics: high marginal fidelity (KS) does not imply high integrity, exactly the pattern observed for Vine Copulas on constrained domains.

4.2 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility, a validation paradigm now standard in privacy-sensitive domains [46]. We measured downstream TSTR F1 under integrity filtering across representative generative architectures (Table 3), using a Random Forest classifier [34] (via *Scikit-learn* [47]); the *Rule-only Baseline* prunes records violating any mined implication rules, whereas the *HIF Filtered* variant uses the semantic frontier ($H < 0.5$).

The paired tests on Census ACS ($N = 10$ seeds, binary median-split `household_income` target) quantify the asymmetry. For the Gaussian Copula, filtering yields $p = 0.90$ (95% CI $[-0.003, +0.003]$): no detectable utility degradation, as expected for a cohort largely free of conditional dependency violations (violation rate $< 1\%$). For CTGAN, downstream utility gains are statistically significant (paired t-test $p = 0.002$; Wilcoxon $p = 0.002$; 95% CI $[+0.038, +0.116]$; paired Cohen's $d \approx 1.08$), with F1 improving from 0.381 ± 0.063 to 0.458 ± 0.096 ($+0.077$), supporting the claim that HIF is particularly useful for architectures that produce joint dependency violations. The *Rule-only Baseline* isolates the symbolic layer: pruning the records that violate mined implication rules ($\approx 7\%$ of the cohort; retention 92.9%) leaves utility essentially unchanged (Census ACS CTGAN: F1 0.367 vs. 0.381 unfiltered; Gaussian Copula: 0.941 vs. 0.942), whereas full HIF filtering raises CTGAN F1 to 0.458 at 62.7% retention. The utility recovery is therefore driven by the LSE/NIC conditional-dependency layers rather than by hard-rule pruning alone. On Online Purchases with the Gaussian Copula (39% violation rate), filtering retains 61.3% of records and F1

Table 1 Cross-Architecture Audit: Statistical Fidelity (KS), Support Overlap (α-Precision, β-Recall), and Joint Dependency Integrity (HIF) across benchmark datasets ($N = 10$ seeds, Mean $\pm$ SD). High marginal fidelity (KS > 0.96) or high support coverage (α-Prec > 0.79) does not imply joint conditional integrity. On datasets with cross-feature constraints, models exhibit up to 100% dependency violation rates. The "Real data (held-out)" row per dataset is a single deterministic reference: the auditor fitted on real rows and scored on a genuine held-out remainder (see Section 4), establishing the framework's own error floor—genuine real data is flagged at 0.0–14.0% across domains.

Generator	KS (↑)	α-Prec (↑)	β-Rec (↑)	HIF (↑)	Viol. Rate (↓)
Supermarket Sales					
Real data (held-out)	—	—	—	0.980	0.0%
Gaussian Copula	0.903 ± 0.006	0.892 ± 0.012	0.918 ± 0.013	**0.955 ± 0.008**	**3.4 ± 0.8%**
Vine Copula	**0.975 ± 0.002**	0.794 ± 0.008	0.866 ± 0.015	0.028 ± 0.005	99.9 ± 0.1%
CTGAN	0.785 ± 0.034	0.628 ± 0.069	0.878 ± 0.020	0.015 ± 0.012	100.0 ± 0.0%
TVAE	0.594 ± 0.055	0.523 ± 0.051	0.618 ± 0.073	0.035 ± 0.005	100.0 ± 0.0%
Online Purchases					
Real data (held-out)	—	—	—	0.918	6.8%
Gaussian Copula	0.753 ± 0.010	0.488 ± 0.013	0.822 ± 0.023	**0.637 ± 0.014**	**38.7 ± 2.0%**
Vine Copula	**0.970 ± 0.004**	0.829 ± 0.018	0.952 ± 0.010	0.271 ± 0.007	85.5 ± 1.0%
CTGAN	0.662 ± 0.064	0.524 ± 0.070	0.868 ± 0.068	0.304 ± 0.042	81.3 ± 5.3%
TVAE	0.718 ± 0.018	0.610 ± 0.033	0.487 ± 0.026	0.149 ± 0.012	99.8 ± 0.1%
Credit Default					
Real data (held-out)	—	—	—	0.823	14.0%
Gaussian Copula	0.749 ± 0.004	0.738 ± 0.024	0.951 ± 0.010	0.487 ± 0.081	46.2 ± 9.9%
Vine Copula	**0.982 ± 0.002**	0.801 ± 0.011	0.813 ± 0.013	0.404 ± 0.039	56.2 ± 4.6%
CTGAN	0.728 ± 0.063	0.672 ± 0.106	0.855 ± 0.028	0.305 ± 0.052	70.3 ± 6.7%
TVAE	0.732 ± 0.015	0.515 ± 0.034	0.762 ± 0.032	**0.908 ± 0.053**	**8.0 ± 5.3%**
Adult Income					
Real data (held-out)	—	—	—	0.923	4.3%
Gaussian Copula	0.854 ± 0.006	0.936 ± 0.007	0.836 ± 0.014	0.795 ± 0.044	14.8 ± 5.2%
Vine Copula	**0.982 ± 0.005**	**0.968 ± 0.011**	0.788 ± 0.016	0.716 ± 0.069	22.1 ± 7.8%
CTGAN	0.692 ± 0.085	0.804 ± 0.034	0.808 ± 0.050	0.618 ± 0.053	34.4 ± 5.9%
TVAE	0.751 ± 0.019	0.350 ± 0.065	0.425 ± 0.042	**0.990 ± 0.007**	**0.2 ± 0.4%**

moves within noise ($0.977 \rightarrow 0.980$), tempering the hypothesis that dependency violations always act as predictive noise: whether removing them improves utility depends on the strength of the violation-to-label association.

The utility-recovery conclusion is robust to the operating threshold, not merely to the continuous score (Table 8). Sweeping the retention frontier $H \geq \{0.7, 0.5, 0.3\}$ on the Census ACS CTGAN cohort ($N = 10$ seeds) yields F1 of 0.761, 0.458, and 0.382 at 30.0%, 62.7%, and 88.5% retention (unfiltered 0.381). The recovery effect is significant at the strict and default frontiers (paired t-test $p \leq 0.002$ and Wilcoxon $p \leq 0.002$; 95% CIs $[+0.332, +0.428]$ and $[+0.038, +0.116]$) and increases monotonically with filtering strictness, but it disappears at the loose $H \geq 0.3$ frontier ($p = 0.92$), where only the most egregious records are removed.

Protocol disclosure. As is standard in TSTR [46], we fit each generator on the full real cohort and evaluate the downstream classifier on a 30% real holdout carved from that same cohort; the synthetic training distribution therefore overlaps the real rows used as test data, and absolute TSTR levels are optimistically biased. This does not affect the paired filtering comparisons reported below, since both arms share the identical synthetic cohort and the ΔF1 differences are protected from the leak; the

Table 2 External Verification of HIF-Flagged Records against Real-Data Arithmetic Identities ($N = 10$ seeds, mean; $\pm$ denotes SEM). Flagged records ($H < 0.5$) are independently confirmed as identity-violating at $\geq 98.9\%$ in every configuration, confirming that the HIF penalty targets genuinely invalid structure. Where both valid and invalid cohorts coexist (Gaussian Copula), unflagged records are markedly less likely to violate (**99.6**% and **71.5**% confirmed vs. 98.9–100% among flagged) and the HIF penalty tracks identity severity (Spearman **0.54** and **0.46**). This audit re-fits HIF on the projection onto the five identity-bearing columns, where the arithmetic identity becomes a near-deterministic hub; its Flagged rate is therefore not directly comparable to the full-feature Viol. Rate of Table 1 (e.g., 32.0% vs. 3.4% for Gaussian Copula on Supermarket Sales, which is fully constrained only in the projection).

Dataset	Generator	Flagged	Base Viol.	Conf. @Flagged	Conf. @Unflagged	Spearman ρ
Supermarket Sales						
	Gaussian Copula	33.1%	99.7%	100.0%	**99.6%**	**0.535 $\pm$ 0.006**
	Vine Copula	99.7%	100.0%	100.0%	100.0%	0.087 $\pm$ 0.010
	CTGAN	99.8%	100.0%	100.0%	100.0%	-0.006 $\pm$ 0.027
	TVAE	90.2%	100.0%	100.0%	100.0%	0.459 $\pm$ 0.006
Online Purchases						
	Gaussian Copula	54.8%	86.7%	98.9%	**72.0%**	**0.465 $\pm$ 0.008**
	Vine Copula	97.2%	100.0%	100.0%	100.0%	0.120 $\pm$ 0.018
	CTGAN	96.5%	99.8%	99.9%	98.4%	0.058 $\pm$ 0.020
	TVAE	100.0%	100.0%	100.0%	100.0%	0.179 $\pm$ 0.011

cross-architecture conclusions of Section 4.1 likewise do not rely on TSTR levels. We disclose the protocol rather than present absolute TSTR levels as unbiased estimates.

To characterize when integrity filtering recovers utility across domains, we tested the paired F1 difference (HIF-filtered minus full synthetic) across all dataset–generator combinations of the cross-architecture benchmark with $N = 10$ seeds. Significant utility recovery occurs in four configurations: Census ACS (ΔF1 $= +0.077$, 95% CI $[+0.038, +0.116]$, $p = 0.002$ for CTGAN; $+0.208$, CI $[+0.187, +0.229]$, $p < 0.001$ for Vine Copula) and Online Purchases (ΔF1 $= +0.141$, CI $[+0.033, +0.249]$, $p = 0.016$ for CTGAN; $+0.385$, CI $[+0.325, +0.445]$, $p < 0.001$ for Vine Copula). Recovery is selective rather than a deterministic consequence of the violation rate: high-violation cohorts on Credit Default show no significant change despite 70.3% and 56.2% violation rates (CTGAN, ΔF1 $= +0.011$, $p = 0.15$; Vine Copula, $+0.004$, $p = 0.06$), and Online Purchases Gaussian Copula shows none at 38.7% ($+0.003$, $p = 0.61$). Whether removing flagged records improves utility therefore depends on the strength of the association between the violated conditional structure and the downstream target, not on the violation rate alone. Section 4.3 makes that dependence precise and shows it to be governed by whether the target is itself one of the auditor's hubs. Conversely, most remaining configurations exhibit small, non-significant changes (e.g., Census ACS Gaussian Copula, ΔF1 $= +0.000$, $p = 0.90$; Adult TVAE, $+0.001$, $p = 0.77$; Adult Gaussian Copula, $+0.002$, $p = 0.77$), confirming that filtering does not penalize well-formed synthetic data. No configuration shows a significant degradation; the nearest is Credit Gaussian Copula (ΔF1 $= -0.020$, CI $[-0.042, +0.002]$, $p = 0.073$), indicating that pruning can discard flagged records that retain predictive signal for the target, but this remains a marginal rather than a significant effect. Finally, four transaction configurations (Supermarket Sales CTGAN/TVAE/Vine; Online Purchases TVAE)

Table 3 Downstream utility filtering across all configurations (Mean ± SD, $N = 10$ seeds). Retained cohort size (Retention) dictates whether utility can be reliably estimated. In every configuration with a significant gain, the median-split target is itself one of the auditor's selected hubs (`household_income` for Census ACS, `item_total` for Online Purchases); Section 4.3 shows this to be the governing condition.

Dataset	Generator	Retention (%)	F1 (Full)	F1 (HIF Filtered)	Δ F1
Adult	CTGAN	65.6	0.444 ± 0.016	0.451 ± 0.018	+0.007
	Gaussian Copula	85.2	0.599 ± 0.025	0.601 ± 0.033	+0.002
	TVAE	99.8	0.499 ± 0.067	0.499 ± 0.067	+0.001
	Vine Copula	77.8	0.466 ± 0.030	0.473 ± 0.032	+0.007
Census ACS	CTGAN	62.7	0.381 ± 0.063	**0.458 ± 0.096**	**+0.077**
	Gaussian Copula	98.7	0.942 ± 0.009	0.942 ± 0.008	+0.000
	TVAE	92.9	0.665 ± 0.149	0.675 ± 0.136	+0.011
	Vine Copula	76.4	0.483 ± 0.062	**0.691 ± 0.035**	**+0.208**
Credit	CTGAN	29.6	0.447 ± 0.012	0.458 ± 0.020	+0.011
	Gaussian Copula	53.8	0.486 ± 0.030	0.466 ± 0.020	-0.020
	TVAE	92.0	0.438 ± 0.003	0.438 ± 0.003	+0.000
	Vine Copula	43.8	0.439 ± 0.006	0.443 ± 0.006	+0.004
Purchases	CTGAN	19.1	0.375 ± 0.078	**0.516 ± 0.196**	**+0.141**
	Gaussian Copula	61.3	0.977 ± 0.009	0.980 ± 0.022	+0.003
	TVAE	0.2	0.690 ± 0.175	n.d.	-
	Vine Copula	13.9	0.450 ± 0.077	**0.835 ± 0.071**	**+0.385**
Supermarket	CTGAN	0.0	0.343 ± 0.023	n.d.	-
	Gaussian Copula	96.6	0.999 ± 0.001	0.999 ± 0.001	+0.000
	TVAE	0.0	0.766 ± 0.153	n.d.	-
	Vine Copula	0.2	0.458 ± 0.097	n.d.	-

* Full-cohort F1 is always valid and is reported; the filtered F1 is undefined (n.d.) where the retained cohort is empty (0%). Bold Δ F1 values are statistically significant (paired t-test, $p < 0.05$).

retain at most ≈ 2% of records, so their filtered F1 is not reliably estimable across seeds and their deltas are not reported; their valid full-cohort F1 is given in Table 3. The Online Purchases TVAE cohort retains essentially no records (0.2%), consistent with a cohort whose full-F1 (0.690 ± 0.175) rests on high-utility but structurally implausible rows; blanket filtering would destroy it, reinforcing the recommendation to reserve exclusionary filtering for applications that tolerate cohort shrinkage (Section 5.2).

Because sixteen dataset–generator configurations are tested at $\alpha = 0.05$, we report a multiple-comparison sensitivity check in addition to the individual paired tests. Under a Bonferroni correction ($\alpha = 0.05/16 \approx 0.003$), only the two Vine Copula recoveries (both $p < 0.001$) remain significant, whereas the Census ACS CTGAN gain ($p = 0.002$) and the Online Purchases CTGAN gain ($p = 0.016$) fall below the corrected threshold and should be read as suggestive rather than conclusive. The aggregate pattern is nevertheless unlikely to arise by chance: four of sixteen tests reach $p < 0.05$ versus roughly one expected under a global null, and the recovered effects concentrate precisely on the cohorts with the strongest dependency violations.

4.3 The Recovery Effect Is Label-Conditional

The four recoveries above share a property that the violation rate does not explain: in each case the median-split prediction target is itself one of the auditor's five selected hubs (`household_income` on Census ACS, `item_total` on Online Purchases). Because a sentinel trained on that hub scores each record by how probable its target value is given the remaining features, low-H records are enriched for rows whose *label* is atypical, not only for rows whose feature geometry is atypical. Pruning them would then improve TSTR F1 through conditional label curation rather than through structural repair. We ran two experiments to determine whether this is what drives the effect.

Experiment 1: varying the target at fixed cohort.

We held the retained cohorts exactly as published and swept the downstream target across nine Census ACS columns—the five selected hubs and four non-hubs—so that the filter, the retained rows, and the retention rate are byte-identical across all conditions and only the label changes (Table 4, Panel A). Nine of the ten hub target–generator pairs recover significantly (the sole exception is CTGAN×tenure, $p = 0.109$), with group-mean gains of +0.077 (CTGAN) and +0.112 (Vine Copula). Seven of the eight non-hub pairs show no detectable effect (the exception is Vine Copula×cost_burden_pct, $p = 0.016$), with group means of +0.014 and +0.010—an order of magnitude smaller. Among the significant pairs every hub confidence interval excludes zero, and among the non-hub pairs only the cost_burden_pct interval does. Because the retained cohort is fixed, this contrast cannot be attributed to which records the auditor removed; it is entirely a property of which column is being predicted.

Notably, `cost_burden_pct` is statistically entangled with two hubs—it correlates 0.72 with $12 \cdot$ `housing_cost/household_income`—yet shows no recovery under CTGAN (+0.020, $p = 0.16$) and only a small marginal one under the Vine Copula (+0.020, $p = 0.016$), an order of magnitude below the hub-driven gains. Proximity to an audited hub is not sufficient; the target must be audited directly.

Experiment 2: withholding the target from the auditor.

We then re-ran the four published recoveries under two interventions (Table 4, Panel B): arm B removes the target from the hub candidate set, so the auditor still sees every column but no sentinel predicts the target; arm C removes the target column from the auditor entirely. Retention rises in every arm, as expected once the target can no longer contribute penalties. Under arm B, the two Census ACS recoveries become non-significant ($p \geq 0.24$) and collapse to within ±0.019 F1 of zero, while the two Online Purchases configurations are attenuated but retain significance (CTGAN: +0.021, $p = 0.027$, an 85% attenuation; Vine: +0.104, $p = 0.001$, a 73% attenuation). Under arm C, no configuration retains a significant gain ($p \geq 0.062$), and the two Census ACS effects collapse to within ±0.007 F1 of zero. The residual arm-B effects on Online Purchases indicate that on that dataset a second pathway contributes—`item_total` participates in arithmetic identities that the NIC and rule layers audit independently of hub selection—so the mechanism is dominant rather than exclusive there.

Interpretation.

Taken together, the two experiments identify a necessary condition rather than a correlate: integrity filtering recovers downstream utility when the auditor models the conditional structure of the target, and not otherwise. We therefore describe contribution (ii) as *label-conditional curation*. This is a narrower claim than generic structural remediation, and it partially overlaps with conditional label-noise cleaning, a connection we make explicit in Section 5.3. It is also more actionable: the precondition is the intersection of the auditor's hub set with the intended prediction target, which a practitioner can check from calibration output alone, before committing to a filtering policy and without consulting downstream labels. The diagnostic contribution (i) is unaffected—HIF's detection of dependency ruptures in Sections 4.1 and 4.4 involves no downstream target at all.

Table 4 The recovery effect is label-conditional. **Panel A**: downstream target swept across nine Census ACS columns at fixed retained cohorts ($N = 10$ seeds; retention 63.3% CTGAN, 76.4% Vine Copula, identical in every row). **Panel B**: the four published recoveries re-run with the target withheld from the auditor—arm A is the published configuration, arm B removes the target from the hub candidate set, arm C hides the target column from the auditor entirely. Bold ΔF1 values are significant at $p < 0.05$ (paired t-test).

Panel A—Target sweep at fixed cohort (Census ACS)

Target	Hub?	CTGAN		Vine Copula	
		ΔF1	p	ΔF1	p
`household_income`	yes	**+0.056**	0.003	**+0.208**	< 0.001
`housing_cost`	yes	**+0.104**	0.002	**+0.114**	< 0.001
`poverty_status`	yes	**+0.091**	0.003	**+0.125**	< 0.001
`education`	yes	**+0.105**	< 0.001	**+0.082**	< 0.001
`tenure`	yes	+0.030	0.109	**+0.030**	0.025
`cost_burden_pct`	no	+0.020	0.16	**+0.020**	0.016
`employment_status`	no	+0.020	0.18	−0.002	0.85
`household_size`	no	+0.020	0.19	+0.014	0.15
`age_group`	no	−0.002	0.77	+0.007	0.45
Hub group mean		+0.077		+0.112	
Non-hub group mean		+0.014		+0.010	

Panel B—Withholding the target from the auditor

Configuration	Arm	Retained (%)	ΔF1	p
Census ACS CTGAN	A	62.7	**+0.077**	0.002
	B	78.2	−0.019	0.24
	C	90.4	−0.003	0.78
Census ACS Vine	A	76.4	**+0.208**	< 0.001
	B	88.3	−0.000	0.98
	C	93.7	+0.007	0.46
Purchases CTGAN	A	19.1	**+0.141**	0.016
	B	30.3	+0.021	0.027
	C	17.6	+0.007	0.68
Purchases Vine	A	13.9	**+0.385**	< 0.001
	B	25.5	+0.104	0.001
	C	15.9	+0.073	0.062

4.4 Comparison with Outlier Detection and Learned-Density Baselines

A natural question is whether standard outlier detection methods—Isolation Forest (IF) [48] and Local Outlier Factor (LOF) [49]—or a learned-density baseline can serve as alternatives to HIF. Table 5 compares detection performance across five error types on Census ACS at 40% corruption ($N = 10$ seeds); Table 6 replicates the identical protocol on Online Purchases. LOF and IF are configured with a `contamination` parameter set to the known corruption rate (0.4), giving them a tuned decision threshold for F1 in this stress-test setup, whereas HIF uses its fixed non-parametric default frontier, flagging records with $H < 0.5$. As a learned-density baseline we fit a Gaussian Mixture Model (GMM) on the real data with component count selected by the Bayesian information criterion [50] and score synthetic records by negative log-likelihood; its F1 threshold is the $(1-0.4)$ quantile of the real-data scores, the density analogue of the oracle `contamination` calibration granted to IF and LOF. For a like-for-like comparison we additionally report HIF† and GMM†, which flag the top 40% of records by their respective scores, matching the calibration granted to the baselines; in practice the baselines flag a comparable fraction of records (IF $\approx$ 40% on average, 35–46% across error families; LOF $\approx$ 79%, 63–87%; GMM $\approx$ 72%, 60–83%), so the matched HIF operating point is not laxer than theirs.

Under threshold-independent ROC-AUC on Census ACS, HIF strongly outperforms the geometric baselines on *Conditional Dependency Violations* (ROC-AUC = 0.824, PR-AUC = 0.776), whereas Isolation Forest is near-random on marginal-preserving dependency swaps (ROC-AUC = 0.514) and LOF achieves lower ranking quality (ROC-AUC = 0.717, PR-AUC = 0.655). A learned-density baseline is competitive with HIF on this family (GMM ROC-AUC = 0.819, PR-AUC = 0.802) but trails HIF at the matched operating point (F1 0.710 vs. 0.719; Table 5, Panel B). The F1 comparison is markedly more favorable to HIF once operating points are matched: flagging the top 40% of records by HIF penalty yields the highest F1 in three of five error families—F1 = 0.719 on dependency violations (vs. 0.594 for LOF, 0.438 for IF, and 0.710 for GMM†), F1 = 0.542 on feature dropout (vs. 0.521 for LOF), and F1 = 0.790 on random noise injection (vs. 0.616 for LOF)—despite flagging only 40% of records to LOF's $\approx$ 86%. HIF also maintains high sensitivity on random noise (ROC-AUC = 0.890) because breaking marginals simultaneously disrupts conditional dependencies. The PR-AUC results are consistent with ROC-AUC across the five families (Table 5, Panel A): HIF leads on dependency violations and feature dropout, and the learned-density baseline is competitive on random noise injection.

Two of the five families are negative controls by construction, so the values in these rows are sanity checks rather than benchmarks. *Row duplication* copies rows drawn from the synthetic cohort itself and perturbs them with $N(0, 0.01\sigma)$ noise on numeric columns, so every planted row is in-distribution and individually valid; detecting it would require a cross-row near-duplicate search, and indeed all four methods sit at chance (ROC-AUC 0.501–0.508 on Census ACS, 0.501–0.511 on Online Purchases). *Covariate shift* replaces rows with verbatim real records sampled from the upper quartile of the first numeric column, which are genuine and *more* plausible than the surrounding synthetic rows, so every plausibility-based scorer lands at or below chance; sub-chance AUC is the correct response to this protocol, not a failure of the method. The apparent wins in these two rows are operating-point artifacts: at the 40% prevalence used throughout, randomly flagging a fraction q of records attains $\mathrm{F1} = 2\pi q/(\pi + q)$ with $\pi = 0.4$, and LOF's row-duplication F1 of 0.535—obtained by flagging 80.3% of records—equals the random expectation of 0.534, just as GMM's 0.544 on Online Purchases (84.3% flag rate) matches its expectation of 0.543. The three informative families are therefore random noise injection, feature dropout, and conditional dependency violations.

The second-dataset replication on Online Purchases (Table 6) confirms the pattern against the geometric detectors—HIF leads IF and LOF on dependency violations (ROC-AUC 0.802 vs. 0.491 and 0.584), feature dropout (0.714 vs. 0.287 and 0.613), and random noise (0.829 vs. 0.349 and 0.521)—but exposes a clear boundary of the learned-density baseline: GMM is markedly stronger on this domain. On dependency violations GMM attains ROC-AUC 0.954 and PR-AUC 0.908 (vs. 0.802 and 0.639 for HIF) and a matched F1 of 0.890 (vs. 0.668), and it also leads on feature dropout (matched F1 0.737 vs. 0.573) and random noise (0.720 vs. 0.690). We attribute this to the dataset's near-deterministic continuous arithmetic identities ($\mathrm{item\_total} = \mathrm{item\_subtotal} + \mathrm{item\_tax}$; $\mathrm{item\_subtotal} = \mathrm{purchase\_price} \times \mathrm{quantity}$): violating these constraints moves a row sharply off a low-dimensional manifold that a Gaussian mixture approximates almost exactly, whereas HIF's categorical-centric sentinel ensemble has little structure to exploit (a single categorical column) and its NIC residual thresholds are conservatively calibrated to the natural noise floor. HIF's advantage is therefore concentrated precisely where learned-density baselines are weakest: predominantly continuous spaces whose conditional structure is complex and non-arithmetic (Census ACS: nine numeric features plus a single 51-level categorical, with all five audited hubs continuous), where the one-hot expansion of that categorical into 51 binary dimensions gives the Gaussian mixture a mostly-binary support on which to score the sharp continuous dependencies.

Why do standard outlier detectors fail on Dependency Gaps? Isolation Forest and LOF evaluate geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD) where both values are common marginally, the record resides in a dense region of the globally projected feature space; because Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A learned-density model faces a related but distinct limitation: on Census ACS it models a 60-dimensional encoding—nine standardized numeric columns joined with 51 one-hot indicators for the sole categorical feature (`state`)—so the sharp conditional structure concentrated in the continuous features is scored through a mostly-binary support, which blurs the contrast between a valid record and a dependency-violating one—yet it still comes close to HIF on this family. A Dependency Gap is a structural, semantic error, not a geometric density anomaly, which is why HIF's logic-aware sentinels are competitive with or better than both geometric and learned-density baselines on conditional dependencies over discrete-valued hubs. The replication further shows this advantage is domain-dependent: when the underlying constraints reduce to continuous arithmetic identities, a learned-density model captures them more sharply than HIF's conservatively calibrated residuals, and the two approaches should be treated as complementary—GMM for near-deterministic continuous constraints, HIF for complex conditional structure over discrete-valued hubs, where it additionally provides per-dependency record attribution that a scalar density score does not.

Panel A: ROC-AUC and PR-AUC (↑)

Table 5 Held-Out Error Detection: HIF vs. Baselines (Census ACS, 40% corruption). Panel A reports threshold-independent ROC-AUC and PR-AUC across 10 seeds; Panel B reports F1 at fixed operating points. HIF† and GMM† flag the top 40% of records by their respective scores, matching the oracle `contamination` calibration granted to IF, LOF, and the GMM quantile threshold; unmarked HIF uses the fixed default frontier, flagging records with $H < 0.5$. At a matched operating point HIF attains the highest F1 on conditional dependency violations (0.719 vs. 0.594 for LOF and 0.710 for GMM†), feature dropout (0.542 vs. 0.521), and random noise injection (0.790 vs. 0.616), while the learned-density baseline is competitive at the level of ranking quality (PR-AUC 0.802 on dependency violations vs. 0.776 for HIF). The row-duplication and covariate-shift rows are negative controls (see text); for these rows the random baselines are ROC-AUC 0.5, PR-AUC ≈ 0.40 (the 40% prevalence), and F1 ≈ 0.40 at a matched 40% flag rate.

	ROC-AUC				**PR-AUC**			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	**0.890**	0.524	0.864	0.874	**0.849**	0.412	0.831	0.848
Row Duplication	0.508	0.504	0.508	0.501	0.408	0.405	0.408	0.404
Feature Dropout	**0.645**	0.315	0.486	0.506	**0.571**	0.295	0.396	0.445
Covariate Shift	0.293	0.469	0.213	0.265	0.355	0.406	0.267	0.280
Conditional Dependency Violations	**0.824**	0.514	0.717	0.819	0.776	0.409	0.655	**0.802**

Panel B: F1 Score at Operating Points (↑)

Error Type	**HIF**	**HIF†**	**IF**	**LOF**	**GMM**	**GMM†**
Random Noise Injection	0.370	**0.790**	0.450	0.616	0.632	0.756
Row Duplication	0.026	0.406	0.423	0.535	0.515	0.402
Feature Dropout	0.072	**0.542**	0.196	0.521	0.492	0.415
Covariate Shift	0.014	0.165	0.387	0.287	0.321	0.221
Conditional Dependency Violations	0.206	**0.719**	0.438	0.594	0.616	0.710

4.5 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \leq 0.1$) using a conditional-swap protocol whose corrupted rows draw replacement feature values from the real conditional pools (Census ACS, $N = 3$ seeds). At these low rates, the HIF score decays monotonically with p (pooled Spearman $\rho = -0.68$ over the $p \leq 0.1$ levels; from 0.933 at $p = 0$ to 0.913 at $p = 0.1$), whereas the aggregate marginal metrics move negligibly—Moment Matching improves only slightly as replacement values are drawn from real conditional pools ($88.56 \rightarrow 88.74$) and KS is essentially unchanged ($0.927 \rightarrow 0.928$)—and the Joint Correlation Distance moves only marginally ($94.6 \rightarrow 94.1$, relative shift

Panel A: ROC-AUC and PR-AUC (↑)

Table 6 Held-Out Error Detection Replication: HIF vs. Baselines (Online Purchases, 40% corruption). Identical protocol and operating-point conventions as Table 5. On this continuously constrained domain the learned-density baseline is markedly stronger than HIF on ranking quality for *Conditional Dependency Violations* (ROC-AUC 0.954 vs. 0.802; matched F1 0.890 vs. 0.668), while HIF retains an edge over the geometric detectors (IF, LOF) on all three dependency-relevant families. As in Table 5, the row-duplication and covariate-shift rows are negative controls (see text).

	ROC-AUC				PR-AUC			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	0.829	0.349	0.521	**0.835**	0.736	0.310	0.458	**0.757**
Row Duplication	0.504	0.501	0.505	0.511	0.409	0.402	0.407	0.409
Feature Dropout	0.714	0.287	0.613	**0.813**	0.570	0.289	0.516	**0.758**
Covariate Shift	0.190	0.294	0.284	0.176	0.341	0.310	0.317	0.258
Conditional Dependency Violations	0.802	0.491	0.584	**0.954**	0.639	0.385	0.479	**0.908**

Panel B: F1 Score at Operating Points (↑)

Error Type	**HIF**	**HIF**†	**IF**	**LOF**	**GMM**	**GMM**†
Random Noise Injection	0.706	0.690	0.397	0.496	0.608	**0.720**
Feature Dropout	0.624	0.575	0.320	0.544	0.592	**0.737**
Covariate Shift	0.012	0.056	0.325	0.297	0.303	0.124
Row Duplication	0.391	0.398	0.509	0.516	0.544	0.413
Conditional Dependency Violations	0.707	0.668	0.511	0.551	0.613	**0.890**

$< 0.6\%$). The small absolute JCD change reflects dilution: the injected rows ($\leq 10\%$ of the 2,000-row cohort) contribute negligibly to a cohort-averaged correlation statistic. Because HIF emits a per-record penalty, it localizes exactly which rows carry the corrupted conditional relationships—a capability no aggregate metric provides. This reading is complementary, not adversarial: under the broad, high-rate permutation protocol described earlier (which destroys pairwise correlations), the JCD decays sharply in lockstep with the HIF score. The two tools therefore address related but distinct failure modes: the JCD is a sensitive dataset-level drift detector, while HIF provides record-level attribution of conditional dependency violations.

4.6 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on Census ACS using CTGAN ($N = 5$ seeds; Table 7). Because Census ACS is predominantly continuous—nine numeric columns plus a single 51-level categorical (`state`), with all five audited hubs quantile-binned continuous features—the Logical Sentinel Ensemble (LSE), which scores conditional structure through these hubs, is the most critical component, raising F1 from 0.412 ± 0.032 (no filtering) to 0.837 ± 0.012 while retaining 26.9% of records. In contrast, rule-based filtering flags few records (7.2% filtered out) and leaves utility near baseline, and NIC-only pruning barely changes retention (7.9% filtered out), indicating that the continuous-residual signal NIC thresholds is secondary to the hub-conditional signal on this domain; combining LSE with NIC yields intermediate behavior (retention 44.5%, F1 0.672 ± 0.058).

The ablation also illustrates the aggregation method's role: Full HIF with multiplicative product aggregation ($F1 = 0.490 \pm 0.043$) outperforms the arithmetic mean ($F1 = 0.419 \pm 0.026$). The multiplicative product acts as a non-compensatory (Logical AND) gate—preventing a high score in one dimension from hiding a severe failure in another—pruning an additional 28% of records relative to arithmetic aggregation (retaining 64.0% vs. 91.8%). The magnitude of this advantage should be read with caution: retention and utility are confounded, and the gap is driven primarily by how aggressively each variant prunes. Consistent with this interpretation, the aggregation choice is immaterial in settings free of structural errors: under Gaussian Copula on Census ACS, where even the most aggressive variant (LSE-only) filters just 4.9% of records, all ablation configurations yield statistically indistinguishable F1 (0.945–0.948).

Why Adding Components Lowers the Flag Rate.

It may seem counterintuitive that adding components (NIC, Rules) to the Logical Sentinel Ensemble (LSE) decreases the flag rate and lowers F1 compared to LSE alone. This behavior stems from aggregating continuous probabilities rather than using boolean rejection. When LSE assigns a harshly low score to a synthetic record, LSE-only will flag it as a violation (leading to a high rejection rate and high F1 on the small remaining cohort). However, if the record is structurally sound in continuous features and rules, NIC and Rule modules will assign it scores near 1.0. Averaging these continuous scores pulls the overall HIF score up, often rescuing the borderline record from falling below the $H < 0.5$ threshold. Consequently, Full HIF acts as a balanced filter that retains a larger, more diverse dataset (yielding an F1 of 0.490 with 64.0% retention) rather than an ultra-strict, low-diversity filter (LSE-only F1 0.837 with 26.9% retention). The geometric mean is used specifically because it is harder to "rescue" a low score via multiplication than arithmetic addition, pruning more aggressively while avoiding total cohort collapse.

Table 7 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds, Mean $\pm$ SEM). The multiplicative product implements a non-compensatory Logical AND gate, pruning more records than the arithmetic mean and reaching the highest F1 on a cohort with structural errors. In cohorts free of structural errors (Gaussian Copula), all configurations are statistically indistinguishable (F1 0.945–0.948).

Ablation	Retention (%)	F1 (mean ± SEM)	Filtered Out (%)
No filtering	100.0%	0.412 ± 0.032	—
LSE-only	26.9%	0.837 ± 0.012	73.1%
NIC-only	92.1%	0.397 ± 0.026	7.9%
Rules-only	92.8%	0.391 ± 0.022	7.2%
LSE + NIC	44.5%	0.672 ± 0.058	55.5%
Full HIF (arith.)	91.8%	0.419 ± 0.026	8.2%
Full HIF (multiplicative)	**64.0%**	**0.490 ± 0.043**	**36.0%**

5 Discussion

The results indicate that the Hybrid Integrity Framework provides a conservative diagnosis in our experimental settings. By employing a multiplicative aggregation model, HIF ensures that a failure in any single semantic dimension results in a low integrity score. This non-compensatory evaluation is important for high-stakes domains, where a single inconsistent record can invalidate an analysis.

5.1 Reconciling Generative Paradigms: Statistical vs. Neural Trade-offs

A central finding of our cross-architecture audit (Table 1) is that the Dependency Gap is not restricted to deep neural black-boxes; it appears as a recurring pathology affecting both statistical and neural generative models in distinct ways. In our runs, statistical copulas preserve integrity well on Census ACS and Supermarket Sales (HIF 0.94–0.96 for the Gaussian Copula) but degrade on Online Purchases and Credit Default (HIF 0.49–0.64), and the designed Vine Copula control—despite the highest marginal fit (KS ≥ 0.970)—fails joint conditional tests as designed (HIF 0.03–0.27 on the transaction domains). Conversely, neural models such as CTGAN capture discrete categorical clusters on demographic data (HIF 0.57–0.62) but struggle with continuous arithmetic identities (HIF 0.02 on Supermarket Sales). This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone, nor from copula rigidity. It is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity—so post-hoc row-level auditing is necessary regardless of whether the underlying generator is statistical or neural.

5.2 Practical Interpretation, Specificity, and Scalability

Because HIF computes the geometric mean of the component validities across L auditing layers, the overall score $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l\right)^{1/L}$ requires simultaneous joint satisfaction of all conditional models—an intentional conservative design for high-stakes auditing. This specificity–sensitivity trade-off is deliberate: HIF prioritizes *dependency specificity*, flagging violations of learned conditional relationships rather than marginal distributional jitter. A generic anomaly detector sensitive to marginal outliers (such as LOF) would produce false positives on valid but rare records; HIF's sensitivity is bounded by the confidence floors learned from the real data distribution.

From a deployment standpoint, integrity filtering is a curation step whose net value must be weighed against retention: in the two strongest recovery settings the retained cohort falls to $\approx$ 15–19% of the synthetic data (Online Purchases Vine 14.5%, CTGAN 18.7%), and utility gains are selective (significant in four of sixteen configurations). Practitioners should therefore treat HIF primarily as a diagnostic to guide generator refinement and reserve exclusionary filtering for applications that tolerate cohort shrinkage, in line with our recommendation in the broader-impact discussion. **Deployment guidance:** Section 4.3 converts the observed selectivity into a decision rule that can be evaluated before any filtering is applied. At the default $H < 0.5$

frontier, expect a utility gain only when the intended prediction target appears in the auditor's selected hub set; when it does not, expect the filter to be utility-neutral at best, and note that no configuration shows a significant degradation in our runs (the nearest, Credit Gaussian Copula, has $p = 0.073$). Because the hub set is produced by calibration on real data alone, this check costs nothing and requires no downstream labels. Two caveats attach to it. First, it is a necessary condition, not a sufficient one: on Credit Default the designated target `default_payment` is not a hub and shows no recovery, consistent with the rule, but substituting its `pay_*` hubs as targets yields significance in only 3 of 10 pairs at baseline F1 ≈ 0.10, so a hub target whose sentinels carry little discriminative signal does not guarantee a gain. Second, when the target is a hub, filtering is partly performing conditional label curation, so a practitioner whose actual goal is label denoising should compare against a dedicated label-noise method rather than adopt HIF filtering on the strength of the F1 gain alone. Retention below $\approx 15\%$ should trigger manual review before deployment regardless.

The LSE architecture is highly scalable in the number of records, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs, and our sweep (Appendix D.1) shows stable performance even at $K = 1$, meaning the hub count can be minimized to reduce the K-multiplier overhead. In our benchmarks, auditing 10,000 synthetic rows required $\approx 0.70\,\mathrm{s}$ and 100,000 rows $\approx 5.70\,\mathrm{s}$ on standard multi-core commodity hardware (Appendix D.3). These measurements vary the row count on a 10-feature benchmark, and the largest feature count in any of our datasets is 24 (Credit); we have not empirically evaluated very wide tables, where the memory overhead of training and storing K Random Forest sentinels is the likely scaling bottleneck and sparse-linear or neuro-inspired models would offer a better trade-off between semantic depth and memory usage.

5.3 Limitations and Failure Regimes

Despite strong results, several limitations remain. First, HIF is not designed as a global covariate-shift detector, and neither are the row-level baselines: no per-record plausibility score detects distribution-level drift, which is the domain of two-sample tests rather than row-level diagnostics (Section 4.4). Second, its advantage over learned-density baselines (e.g., GMM) is domain-dependent: on datasets whose constraints reduce to near-deterministic continuous arithmetic identities, a density model captures violations more sharply than HIF's conservatively calibrated residuals, and practitioners with such domains should prefer it. Third, the quality of hub selection and confidence-floor calibration depends on sample size and class balance; in sparse regimes, rare-but-valid records may overlap with high-penalty regions. Fourth, the fixed frontier ($H < 0.5$) is practical but not universally optimal; deployment settings may require task-specific threshold calibration and cost-aware tuning, and extending calibration to non-exchangeable data streams via conformal methods [51] remains open. Fifth, the current empirical benchmark set, while diverse, does not exhaust domain types (for example, temporal or relational tables), so external validity should be interpreted cautiously.

Sixth, and most consequential for how the utility result should be read: because the recovery effect requires the prediction target to be among the audited hubs

(Section 4.3), integrity filtering in those settings overlaps functionally with conditional label-noise cleaning. A sentinel predicting the target from the remaining features is, by construction, a conditional label model, and pruning records it scores as improbable removes rows whose target is atypical given their features. We do not claim HIF is preferable to purpose-built label-noise methods on that task, and we have not benchmarked it against them; the distinction we do claim is one of scope—HIF audits all K hubs plus the continuous and rule layers simultaneously, is calibrated without reference to any downstream task, and yields per-dependency attribution rather than a per-row label-noise probability. A direct comparison against conditional label-noise baselines on the hub-target configurations is the natural next step and would sharpen the boundary between the two. The diagnostic results (Sections 4.1 and 4.4) are independent of this limitation, as they involve no downstream target.

Seventh, HIF is a necessary-but-not-sufficient integrity signal and must be read together with a support-coverage metric. A degenerate generator that emits a single real row repeatedly scores perfectly under HIF on three of our five benchmarks—2,000 copies of one row receive $H = 1.000$ with 0.0% violations on Census ACS, Adult, and Credit Default—because a genuine row satisfies the learned conditional structure by construction, and the auditor is blind to the cohort's collapsed support. The same construction is caught at 100% violations on the two transaction benchmarks, since the duplicated row inevitably violates at least one of the mined arithmetic implication rules; the rule layer therefore partially mitigates mode collapse wherever deterministic constraints exist, but it cannot rescue the demographic and financial benchmarks above. High HIF scores must accordingly be interpreted against this ceiling: TVAE's Adult cohort carries the highest HIF of any neural configuration (0.990), yet its α-Precision of 0.350 reveals a support far from the real manifold (Table 1). We therefore recommend reporting HIF jointly with a support-coverage diagnostic such as α-Precision and with generator-side diversity measures, rather than relying on the integrity frontier alone to certify cohort diversity.

5.4 Broader Impact

By providing a verifiable integrity signal, HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using HIF primarily as a diagnostic tool to guide generative model refinement rather than as a purely exclusionary filter.

6 Conclusion

As generative models grow in complexity, the metrics used to evaluate synthetic tabular data must evolve beyond aggregate marginal alignment to assess row-level joint distributional consistency. We introduced the Hybrid Integrity Framework (HIF), a probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data, combining the Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature plausibility within the learned categorical context.

Our empirical results on Census ACS and transaction benchmarks show monotonic HIF degradation with increasing targeted dependency corruption in our experiments, while marginal metrics such as Moment Matching and KS remain static under the same corruption; joint correlation metrics also respond to broad corruption, but only HIF attributes it to specific records. Our cross-architecture audit reveals a persistent "Integrity-Fidelity Decoupling": a designed Vine Copula control, which samples categorical columns independently from their marginal distributions, achieves high distributional fidelity (KS > 0.96) while failing 85–100% of dependency integrity tests on constrained datasets. By implementing non-compensatory probabilistic aggregation across independently learned dependency models, HIF provides a conservative row-level signal of joint distributional consistency—conformity to the conditional feature dependencies of the real data—that aggregate marginal metrics often miss. We further show that HIF outperforms standard geometric outlier detectors (Isolation Forest, LOF) on conditional dependency violations while remaining appropriately insensitive to marginal noise that does not constitute a dependency failure, and that this pattern replicates on a second dataset. A learned-density baseline (GMM) is competitive with HIF on the predominantly continuous Census ACS and surpasses it on continuously constrained transaction domains, so HIF's distinctive value is concentrated in spaces whose conditional structure is complex and does not reduce to continuous arithmetic identities, where it additionally provides interpretable per-dependency attribution that a scalar density score does not. On the remediation side, we find that pruning low-integrity records recovers downstream utility under a specific and checkable condition—that the prediction target is among the audited dependency hubs—and that the effect disappears when it is not, whether the target is varied at fixed cohort or withheld from the auditor. Integrity filtering is therefore best understood as label-conditional curation rather than generic structural repair, which narrows the claim but makes it operational: the precondition can be read off the auditor's calibration output before any filtering decision is made. Future work will extend this framework to high-frequency financial time-series data [52], benchmark the hub-target filtering regime against dedicated conditional label-noise methods, and explore kernel-based projections to capture higher-order dependencies.

Statements and Declarations

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Materials availability. Not applicable.

Code availability. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

Data availability. The experimental benchmarks and dataset configurations, together with the complete open-source implementation, are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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Generative AI. During the preparation of this work the authors used AI-assisted tools for drafting and editorial assistance. After using these tools, the authors reviewed and verified the content and take full responsibility for the content of the publication.

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A Formal Results for the HIF Score Under Stated Assumptions

We provide a formal analysis of Hybrid Integrity Framework (HIF) score properties under the assumptions stated below. The results should be interpreted as assumption-scoped statements rather than unconditional guarantees.

None of the statements below is a conformal coverage guarantee. The confidence floor δ_h is the empirical 5th percentile of each sentinel's out-of-bag conditional probabilities, not a conformal quantile computed on an exchangeable calibration set: out-of-bag predictions are not exchangeable with test points in the required sense, and a per-hub *marginal* quantile does not yield a *per-record* bound. HIF therefore provides measured—not guaranteed—operating characteristics; on genuine held-out real data its false-rejection (violation) rate ranges from 0.0% (Supermarket Sales) to 14.0% (Credit Default) across datasets (Section 4).

Proposition 1 (Intersection Soundness). The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$. Under the idealization that each component validity satisfies $\mathcal{C}_l(\mathbf{x}) = 1$ iff $\mathbf{x} \in \mathcal{M}_l$ and

$\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise, $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l(\mathbf{x})\right)^{1/L}$ attains its maximum value 1 exactly on $\mathcal{M}_{total}$:

$$\mathbb{1}[H(\mathbf{x}) = 1] = \prod_l \mathbb{1}[\mathbf{x} \in \mathcal{M}_l] \tag{10}$$

Proof. *Completeness:* if $\mathbf{x} \in \mathcal{M}_{total}$ then $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, so $H(\mathbf{x}) = 1$. *Soundness (non-compensatory property):* if $\mathbf{x} \notin \mathcal{M}_{total}$, there exists l^* with $\mathcal{C}_{l^*}(\mathbf{x}) < 1$, and since all validities lie in $[0, 1]$, the geometric mean satisfies $H(\mathbf{x}) < 1$. *Strict convergence:* an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1$) drives $H(\mathbf{x}) \to 0$ regardless of other layers. $H(\mathbf{x})$ thus behaves as a conservative diagnostic for joint satisfaction of latent manifold constraints, matching the implementation in Algorithm 2.

Proposition 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (the Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$, and the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges pointwise in probability to the corresponding population penalty:

$$\lim_{N \to \infty} P\left(\left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0,\ \mathbf{x} \in \mathcal{M}_{real} \tag{11}$$

Proof sketch. *(i)* Because the LSE leverages consistent non-parametric estimators, the empirical conditional probabilities converge pointwise in probability to the true Bayesian posterior P^* (provided the feature space is bounded and leaf nodes grow sub-linearly with N); the pointwise statement reflects that sup-norm convergence is not generally guaranteed for forest-based estimators. *(ii)* Defining the confidence floor δ_h as the empirical 5th-percentile of the sentinel's out-of-bag probabilities yields an asymptotic upper bound of approximately $\alpha = 0.05$ on false positives under calibration assumptions. *(iii)* Records penalized by the converged HIF ($\mathcal{P}^* > 0$) are expected to lie in lower-probability conditional regions of the manifold, supporting a structural-violation interpretation. *(iv)* By the Uniform Law of Large Numbers and the Continuous Mapping Theorem, the empirical Feature Dependency Score $S^{(N)}(h)$ converges in probability to the Bayes optimal classification accuracy $S^*(h)$, implying asymptotically stable hub ranking.

Remark 1 (Type I Error Bound & Cohort-Purity Approximation). Let $\mathbf{x} \sim P_{real}$ be a valid record on the ground-truth manifold $\mathcal{M}_{real}$ (including rare tail outliers). Under quantile calibration ($\delta_h = Q_\alpha$ and continuous tolerance bandwidth $\gamma_y = Q_{1-\beta}$), the false-rejection probability is upper-bounded by the Union Bound:

$$P\left(H(\mathbf{x}) < 0.5 \mid \mathbf{x} \in \mathcal{M}_{real}\right) \leq \sum_{h \in \mathcal{H}} \alpha_h + \sum_y \beta_y + \gamma_{rule} \tag{12}$$

where γ_{rule} is the rule-layer false-rejection rate of valid records under hard application of the mined implication rules (Section 3); it is nonzero in practice because each rule holds on only $\geq 95\%$ of real records by construction, and its empirical counterpart is the held-out violation rate quantified at the end of this remark. Furthermore, let a generative model produce a noisy mixture $P_{syn} = (1 - \eta)P_{real} + \eta P_{hallucinated}$, where $\eta \in [0,1]$ is the structural hallucination rate. Rejection filtering with threshold $H \geq 0.5$ yields an approximate retained-cohort purity:

$$\text{Purity}(H \geq 0.5) = \frac{(1-\eta)(1-\alpha)}{(1-\eta)(1-\alpha) + \eta\beta_{hallucination}} \tag{13}$$

where $\beta_{hallucination}$ is the leakage probability of hallucinated records past the filter. Equation (13) is a stylized Bayes calculation: it illustrates the mechanism by which filtering can raise retained-cohort purity, but in practice α, β, and η are unknown and must be estimated empirically rather than assumed. The measured counterpart is the false-rejection (violation) rate of HIF on genuine held-out real data, which ranges from 0.0% (Supermarket Sales), 1.3% (Census ACS), and 4.3% (Adult) to 6.8% (Online Purchases) and 14.0% (Credit Default)—see Section 4. The framework is therefore best read as a diagnostic whose operating point is established empirically, not as one carrying guaranteed error bounds.

B End-to-End Auditing Procedure

Algorithm 1 summarizes the complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—and Algorithm 2 details the end-to-end auditing procedure for scoring a synthetic dataset with the trained HIF components.

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

$\mathcal{H} \leftarrow$ Select top K_{hub} features by Feature Dependency Score $S(h)$
for each hub $h \in \mathcal{H}$ **do**
  Train Sentinel $f_h : \mathcal{X}_{\setminus h} \to [0,1]^{|\mathcal{X}_h|}$ on $\mathcal{D}_{real}$ to predict x_h
  $\delta_h \leftarrow \max(\text{Percentile}_5(\{f_h^{\text{OOB}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01)$
end for
$\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Non-Centered Spectral Manifold
for each continuous feature y **do**
  Train Gradient Boosting regressor $r_y : \mathcal{Z} \to \mathbb{R}$ on $\mathcal{D}_{real}$
  $\mathcal{E}_y \leftarrow \{|y - r_y(\mathbf{z})| \mid \mathbf{x} \in \mathcal{D}_{real}\}$ ▷ Ground-truth residuals
  $\tau_y \leftarrow \max(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y))$ ▷ Robust Threshold
  $\gamma_y \leftarrow \max(Q_{0.98}(\mathcal{E}_y), 2 \cdot \tau_y, 3 \cdot \text{MAD}(\mathcal{E}_y), 0.1)$ ▷ Dynamic Gamma Scaling
end for
$\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ Apriori-style, min_conf=0.95, min_supp=0.005

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1
Output: Record Integrity Score $H(\mathbf{x})$
Step 1: Logical Sentinel Audit (Categorical)
for each hub $h \in \mathcal{H}$ **do**
  $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h|\mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$
end for
$\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}}(1 - \mathcal{P}_h)$ ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
$\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$ ▷ Apply learned standardized projection
for each continuous feature y **do**
  $\rho_y \leftarrow |y - r_y(\mathbf{z})|$
  $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$
end for
$\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
Step 3: Integrity Aggregation
$\mathcal{P}_{rule} \leftarrow 1$ if $\mathbf{x}$ violates any $r \in \mathcal{R}$ else 0
$H(\mathbf{x}) \leftarrow \left(\prod_l (1 - \mathcal{P}_l)\right)^{1/L} \in [0, 1]$ ▷ Geometric mean of validities

C Logical Sentinel Ensemble (LSE) Implementation Details

On the U.S. Census ACS dataset, the Feature Dependency Score $S(h)$ identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `tenure`. The confidence floor δ_h is the 5th-percentile of the *out-of-bag* probability mass that a sentinel assigns to the correct category in the ground-truth data (safety floor 0.01); out-of-bag predictions yield out-of-sample floors without a held-out split, reflecting generalization rather than memorized fit, and avoid the instability of 1st-percentile thresholds in sparse data [6, 53]. Hub selection is deterministic: sentinel discovery uses a fixed random state with non-shuffled cross-validation folds, so the hub set depends only on the ground-truth data and is verified identical across all $N = 10$ held-out seeds for every benchmark dataset (e.g., `household_income` through `tenure` on Census ACS; `quantity` through `item_tax` on Online Purchases).

D Extended Results and Sensitivity Analysis

D.1 Hyperparameter Sensitivity

We evaluated HIF robustness across its three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations; $N = 5$ seeds, mean $\pm$ SEM). The continuous HIF score is the primary metric; the violation rate derives from applying the violation threshold to row-level penalties. The HIF score is stable across hub counts (varying by $< 1\%$ from 0.978 ± 0.0004 at $K = 1$ to 0.987 ± 0.0008 at $K \geq 10$), since geometric-mean aggregation prevents added sentinels from diluting the score; the hub set saturates at $K = 10$ because Census ACS contains only 10 features. Scores are likewise stable for confidence-floor percentiles 1–5 (0.9848 ± 0.0003 at percentile 1 vs. 0.9840 ± 0.0003 at percentile 5; minor 0.3% drop to 0.9815 ± 0.0004 at the 10th), making the 5th-percentile default a good sensitivity–robustness balance. Finally, the continuous score is invariant to the binary frontier (0.984 ± 0.0003 at every setting, since the frontier is applied only after scoring), while the violation rate decreases monotonically with the frontier (0.82% at $H < 0.3$, 0.43% at $H < 0.5$, 0.25% at $H < 0.7$), letting practitioners tune the operating point without retraining. The same frontier sweep extends to the downstream-utility protocol (Section 4.2): the integrity-filtered F1 gain on Census ACS CTGAN is significant at the strict and default frontiers and grows monotonically with strictness (Table 8).

Table 8 Downstream-Utility Threshold Sensitivity (Census ACS, CTGAN, $N = 10$ seeds, mean). Integrity-filtered TSTR F1 and retention at each retention frontier; records with H below the frontier are pruned, so a higher frontier is stricter. The unfiltered cohort achieves F1 = 0.381. The recovery effect is significant at the strict and default frontiers ($H \geq 0.5$) and grows with filtering strictness, but vanishes at the loose frontier.

Retention frontier H	Retained	F1	ΔF1	Paired t p	Wilcoxon p	95% CI for ΔF1
$H \geq 0.7$ (strict)	30.0%	0.761	+0.380	< 0.001	0.002	[+0.332, +0.428]
$H \geq 0.5$ (default)	63.0%	0.441	+0.060	0.008	0.002	[+0.020, +0.099]
$H \geq 0.3$ (loose)	88.5%	0.382	+0.001	0.92	0.74	[−0.025, +0.028]

D.2 Sample-Complexity Floor at HIF's Operating Configuration

These bounds are computable at HIF's operating configuration. The identity/uniformity-testing bound $m \gtrsim \sqrt{S}/\varepsilon^2$ [21, 23] inverted at the per-hub, per-category sample sizes HIF actually conditions on yields per-cell detectability floors $\varepsilon_{\text{cell}} = (S_{\text{cell}}/n_{\text{cell}}^2)^{1/4}$, where n_{cell} is the number of real rows with hub value c and S_{cell} is the distinct joint configurations of the non-hub features in that cell. Table 9 reports these values for each benchmark dataset using the auditor's default configuration (5 hubs, 10 quantile bins for continuous features). The cohort-level analogue $\varepsilon_{\text{joint}} = (S_{\text{joint}}/n^2)^{1/4}$ with S_{joint} distinct full-row configurations and n the dataset's own cohort size (2000 for Census ACS, Adult, and Credit; 664 for Online Purchases; 1000 for Supermarket Sales) is also shown.

The worst-case floors ($\varepsilon \approx 1.0$) occur where long-tail hub categories leave cells with $n_{\text{cell}} \leq 12$ (Adult `native_country`, Credit `pay_*`, Online Purchases `quantity`). Census ACS, whose hubs (`household_income`, `housing_cost`, `poverty_status`, `education`, `tenure`) all have balanced category counts ($n_{\text{cell}} \geq 198$), achieves $\varepsilon \approx 0.27$; Supermarket Sales' hubs are likewise balanced ($n_{\text{cell}} = 100$, $\varepsilon \approx 0.32$). The cohort-level analogue $\varepsilon_{\text{joint}} \approx 0.15$–$0.19$ is tighter because the full sample $n = 2000$ is used without splitting, but it delivers no per-record attribution. HIF's row-level design therefore trades per-cell detectability for granularity; the multi-hub geometric aggregation and

Table 9 Sample-Complexity Floor: per-cell detectability $\varepsilon_{\text{cell}}$ inverted from the identity-testing bound at HIF's actual hub-conditioned sample sizes (5 hubs per dataset, 10 quantile bins). The worst cell (smallest n_{cell}) drives the floor; aggregate $\varepsilon_{\text{joint}}$ assumes a global uniformity test on the full joint.

Dataset	Rows	Med. n_{cell}	Min n_{cell}	Med. $\varepsilon_{\text{cell}}$	Worst $\varepsilon_{\text{cell}}$	$\varepsilon_{\text{joint}}$
Census ACS	2000	200	198	0.266	0.267	0.150
Adult	2000	5.5	1	0.654	1.000	0.149
Credit	2000	12	1	0.537	1.000	0.150
Online Purchases	664	66	1	0.325	1.000	0.193
Supermarket Sales	1000	100	100	0.316	0.316	0.178

the confidence-floor safety minimum (0.01) are the design elements that dampen the long-tail cells' influence on the final score.

D.3 Scalability and Baseline Comparisons

We measured HIF's auditing cost directly and compare it analytically—not empirically—against the **SHAP Distance** semantic fidelity metric [9], a representative explainability-aware evaluation baseline; we did not run SHAP Distance ourselves, so no accuracy comparison between the two is claimed here. HIF scoring scales linearly in the number of audited records, $O(N \cdot K)$, where K is the number of hubs. In our empirical scalability benchmark, fitting the 5-hub LSE auditor on 2,000 real records took 9.7 s, after which scoring 10,000 synthetic records took 1.50 ± 0.15 s and scoring 100,000 records took 10.6 ± 0.5 s on commodity multi-core hardware (fit amortized once; wall-clock averaged over three repeats). The marginal per-10,000-row cost at the 100,000-row scale ($10.6 / 10 \approx 1.1$ s per 10k rows) confirms the linear scaling. SHAP-based auditing carries a per-record model-explanation cost rather than a single forward pass: exact Shapley values over D features require $O(2^D)$ coalitions in general, and although TreeSHAP computes them exactly in polynomial time for tree ensembles [54], the per-record cost still scales with ensemble size and depth. HIF's per-record cost is therefore substantially lower at fixed D, which is the basis on which we prefer it for repeated auditing at scale; establishing whether it also matches SHAP Distance on detection accuracy would require a direct empirical comparison that we leave to future work.